The Journal of Open Source Software

Gauss-Radau Small-body Simulator (GRSS): An Open-Source Library for Planetary Defense

Rahil Makadia[1¶], **Davide Farnocchia**[2], **Steven R. Chesley**[2], and **Siegfried Eggl**[1]

1 Department of Aerospace Engineering, University of Illinois at Urbana-Champaign, Urbana, IL 61801, USA **2** Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA 91109, USA ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 11 November 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

In partnership with

AMERICAN ASTRONOMICAL SOCIETY

This article and software are linked with research article DOI 10.3847/PSJ/xxxxx, published in the Planetary Science Journal.

DRAFT

Statement of Need

Modeling the motion of small solar system bodies is of utmost importance when looking at the problem in the context of planetary defense. Reliably computing the orbit of an asteroid or a comet from various observations, and then predicting its trajectory in the future is critical in assessing the associated Earth-impact hazard. The NASA Center for Near-Earth Object Studies at the Jet Propulsion Laboratory (JPL) has developed a suite of state-of-the-art tools over the course of decades for this specific purpose. However, these tools are not publicly available. With the expected increase in the number of Near-Earth Object observations as well as discoveries when new observatories such as the Vera C. Rubin Observatory come online (Schwamb et al., 2023), there is a need for a publicly available library that can reliably perform both orbit determination and propagation for small bodies in the solar system. Such a publicly available library will enable community efforts in planetary defense research.

Summary

In this paper, we present `GRSS`, the Gauss-Radau Small-body Simulator, an open-source library for orbit determination and propagation of small bodies in the solar system. `GRSS` is an open-source software library with a C++11 foundation and a Python binding. The propagator is based on the `IAS15` algorithm, a 15th order integrator based on Gauss-Radau quadrature (Rein & Spiegel, 2014). Only the particles of interest are integrated within `GRSS` to reduce computational cost. The states for the planets and 16 largest main-belt asteroids are computed using memory-mapped JPL digital ephemeris kernels (Park et al., 2021) as done in the `ASSIST` orbit propagator library (Holman et al., 2023). In addition to the propagator, the C++ portion of the library also has the ability to predict impacts and calculate close encounter circumstances using various formulations of the B-plane (Chodas & Yeomans, 1999; Farnocchia et al., 2019; Kizner, 1961; Milani & Valsecchi, 1999; Öpik, 1976).

The C++ functionality is exposed to Python through a binding generated using `pybind11`. The Python layer of `GRSS` uses the propagator as the foundation to compute the orbits of small bodies from a given set of optical and/or radar astrometry from the Minor Planet Center, the JPL Small Body Radar Astrometry database, and the Gaia Focused Product Release solar system observations database. Additionally, the orbit determination modules also have the ability to fit especially demanding measurements such as stellar occultations. These capabilities of the `GRSS` library have already been used to study the the heliocentric changes in the orbit of the (65803) Didymos binary asteroid system as a result of the DART impact (Makadia et al., 2022; Makadia, Chesley, et al., 2024) and for analyzing the impact locations of two asteroids, 2024 BX1 and 2024 RW1 (Makadia, Farnocchia, et al., 2024).

GRSS will continue to be developed in the future, with anticipated contributions including the ability to perform mission studies for future asteroid deflections. GRSS is publicly available to the community through the Python Package Index and the source code is available on GitHub. Therefore, GRSS is a reliable and efficient tool that the community has access to for studying the dynamics of small bodies in the solar system.

Acknowledgements

R.M. acknowledges funding from a NASA Space Technology Graduate Research Opportunities award, NASA contract No. 80NSSC22K1173. The work of S.R.C. and D.F. was carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration (No. 80NM0018D0004). S.E. acknowledges support by the National Science Foundation through Award AST-2307570. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation. This work has made use of data from the European Space Agency mission Gaia, processed by the Gaia Data Processing and Analysis Consortium. This library has also extensively used data and services provided by the International Astronomical Union Minor Planet Center (MPC). Data from the MPC's database is made freely available to the public. Funding for this data and the MPC's operations comes from a NASA Planetary Defense Coordination Office grant (80NSSC22M0024), administered via a University of Maryland - Smithsonian Astrophysical Observatory subaward (106075-Z6415201). The MPC's computing equipment is funded in part by the above award, and in part by funding from the Tamkin Foundation.

References

Chodas, P. W., & Yeomans, D. K. (1999). Orbit determination and estimation of impact probability for near earth objects. *AAS Guidance and Control Conference, Breckenridge, Colorado, USA*. https://hdl.handle.net/2014/16816

Farnocchia, D., Eggl, S., Chodas, P. W., Giorgini, J. D., & Chesley, S. R. (2019). Planetary encounter analysis on the b-plane: A comprehensive formulation. *Celestial Mechanics and Dynamical Astronomy*, *131*(8). https://doi.org/10.1007/s10569-019-9914-4

Holman, M. J., Akmal, A., Farnocchia, D., Rein, H., Payne, M. J., Weryk, R., Tamayo, D., & Hernandez, D. M. (2023). ASSIST: An ephemeris-quality test-particle integrator. *The Planetary Science Journal*, *4*(4), 69. https://doi.org/10.3847/PSJ/acc9a9

Kizner, W. (1961). A Method of Describing Miss Distances for Lunar and Interplanetary Trajectories. *Planetary and Space Science*, *7*, 125–131. https://doi.org/10.1016/0032-0633(61)90293-8

Makadia, R., Chesley, S. R., Farnocchia, D., Naidu, S. P., Souami, D., Tanga, P., Tsiganis, K., Hirabayashi, M., & Eggl, S. (2024). Measurability of the Heliocentric Momentum Enhancement from a Kinetic Impact: The Double Asteroid Redirection Test (DART) Mission. *The Planetary Science Journal*, *5*(2), 38. https://doi.org/10.3847/PSJ/ad1bce

Makadia, R., Farnocchia, D., Chesley, S. R., & Eggl, S. (2024). Gauss-Radau Small-body Simulator (GRSS): An Open-Source Library for Planetary Defense. *The Planetary Science Journal*, *Submitted*.

Makadia, R., Raducan, S. D., Fahnestock, E. G., & Eggl, S. (2022). Heliocentric effects of the DART mission on the (65803) didymos binary asteroid system. *The Planetary Science Journal*, *3*(8), 184. https://doi.org/10.3847/PSJ/ac7de7

Milani, A., & Valsecchi, G. B. (1999). The asteroid identification problem: II. Target plane confidence boundaries. *Icarus*, *140*(2), 408–423. https://doi.org/10.1006/icar.1999.6135

DRAFT

Öpik, E. J. (1976). *Interplanetary encounters: Close-range gravitational interactions*. Elsevier Scientific Pub. Co. ISBN: 0444413715

Park, R. S., Folkner, W. M., Williams, J. G., & Boggs, D. H. (2021). The JPL planetary and lunar ephemerides DE440 and DE441. *The Astronomical Journal*, *161*(3), 105. https://doi.org/10.3847/1538-3881/abd414

Rein, H., & Spiegel, D. S. (2014). ias15: a fast, adaptive, high-order integrator for gravitational dynamics, accurate to machine precision over a billion orbits. *Monthly Notices of the Royal Astronomical Society*, *446*(2), 1424–1437. https://doi.org/10.1093/mnras/stu2164

Schwamb, M. E., Jones, R. L., Yoachim, P., Volk, K., Dorsey, R. C., Opitom, C., Greenstreet, S., Lister, T., Snodgrass, C., Bolin, B. T., Inno, L., Bannister, M. T., Eggl, S., Solontoi, M., Kelley, M. S. P., Jurić, M., Lin, H. W., Ragozzine, D., Bernardinelli, P. H., … Ye, Q. (2023). Tuning the legacy survey of space and time (LSST) observing strategy for solar system science. *The Astrophysical Journal Supplement Series*, *266*(2), 22. https://doi.org/10.3847/1538-4365/acc173

DRAFT